E – LEARNING PLATFORM FOR HEARING IMPAIRED STUDENTS

Introduction: E-Learning Platform for Hearing Impaired Students

- Hearing-impaired students face challenges in conventional e-learning due to reliance on auditory content.
- AI-driven tools now enable more personalized and accessible learning experiences.
- Our system aims to provide an inclusive platform with sign language support, emotion tracking, and adaptive feedback.
- Focus is on creating engaging, real-time, and culturally relevant learning for hearing-impaired learners in Sri Lanka.

Introduction to the Problem

- Hearing-impaired students in Sri Lanka face barriers in accessing inclusive digital education.
- Lack of real-time feedback, cultural relevance, and multisensory support reduces learning effectiveness and student confidence.
- A comprehensive solution is needed to integrate **Sign Language Translation,** Gesture Feedback & Avatars, **Emotion Monitoring, Lip Reading, and Emotion-Aware Music** for a fully inclusive learning experience.

Social or research impact of the solution…

- Enhances accessibility for hearing-impaired communities in Sri Lanka.
- Contributes to low-resource language AI development (Sinhala).
- Opens new opportunities in education, healthcare, and public communication.
- Supports future research on multimodal AI (visual + auditory processing).
- Promotes inclusive technology design with real-world applications.

Previous Research and Its Shortcomings

- Insufficient Support for Sinhala Lip Reading
- Lack of Emotion Tracking
- Limited Music Accessibility
- Limited Interaction in Existing Tools

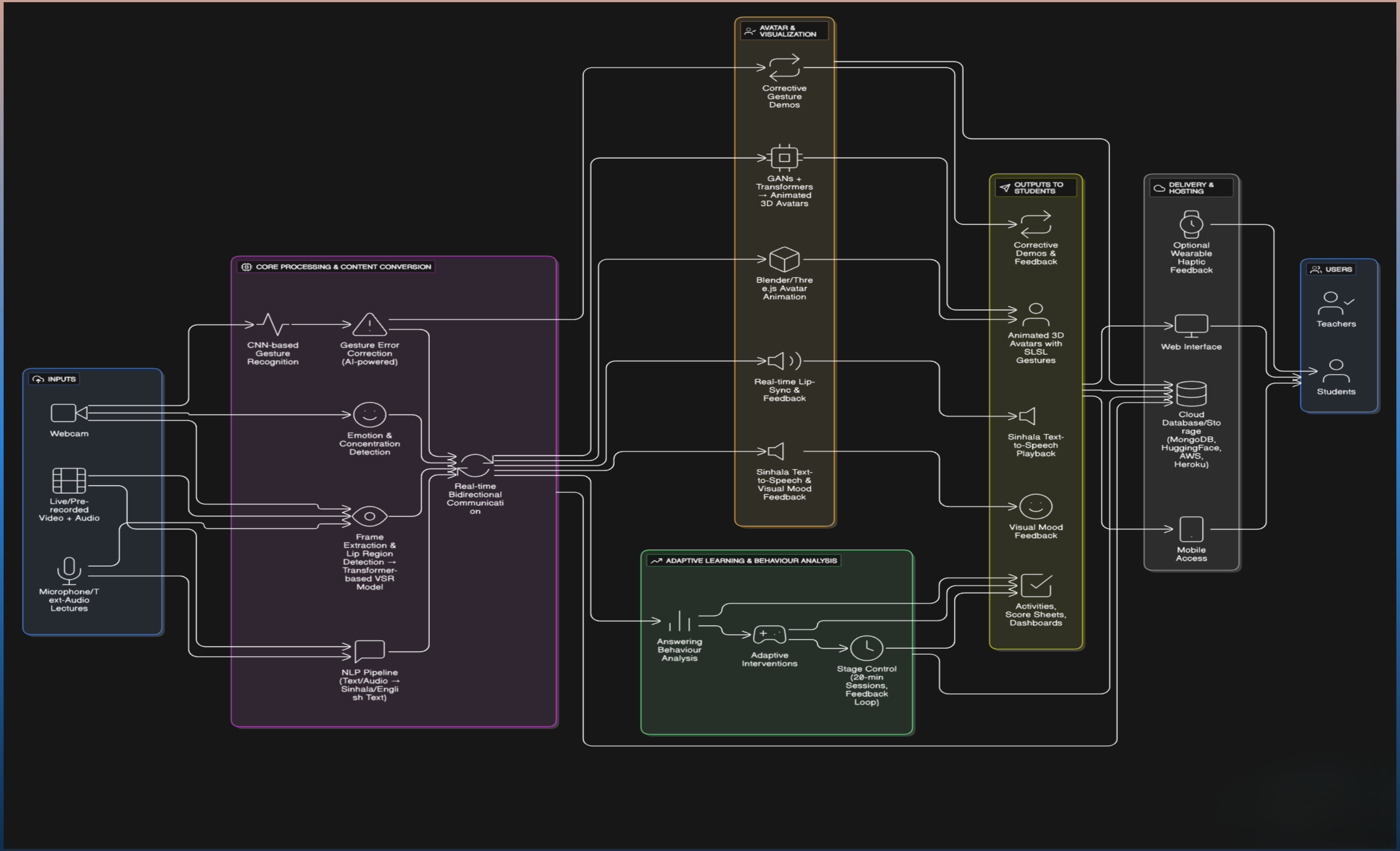
Our Approach to Address These Shortcomings:

- Real-time Bidirectional Communication and customized avatar styles
- Tailored for Sinhala Lip Reading
- Emotion Tracking and Adaptive Learning
- Emotion-Aware Music Experience

Constraints and Limitations

- **Data Availability**: Lack of large-scale, publicly available datasets, especially for Sri Lankan Sign Language (SLSL) and Sinhala lip-reading.
- **Real-time Processing**: High computational power is required for real-time video generation, emotion tracking, and lip-reading assistance.
- **Accuracy**: Emotion recognition depends heavily on facial expressions and environmental factors such as lighting and camera quality.
- **Hardware Compatibility**: Requires devices like wearables, smartphones, and sensors, which may not be accessible to all students.
- **Feedback Synchronization**: Precise synchronization of visual, auditory, and haptic feedback is challenging and requires minimal latency.
- **User Adoption and Training**: Requires extensive user training and adaptation to new technology, which may slow down adoption, especially in regions with lower technical literacy.
- **Computational Resources**: Lip-reading models and emotion tracking require significant computational resources, which may not be available on standard devices.
- **Financial Constraints**: High development and ongoing maintenance costs for advanced technologies, which may impact long-term sustainability and scalability.

System Architecture Diagram



SLIIT FACULTY OF COMPUTING

EMOTION AWARE MUSIC FOR HEARING IMPAIRED STUDENTS

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INTRODUCTION TO INDIVIDUAL COMPONENT

Proven Gap / Creative Solution

- **Gap**: Existing solutions for music accessibility for hearing-impaired students fail to provide an emotionally immersive experience. Most systems rely on basic visual feedback and fail to capture emotional tones or rhythm in an engaging way.
- **Solution**: The proposed system integrates emotion-aware music recognition, haptic feedback, and Sri Lankan Sign Language (SLSL) to create a multisensory, inclusive music education tool for hearing-impaired students.

Knowledge Gap / Problem Definition

- **Gap**: No current solution offers a synchronized multisensory music experience, especially for Sinhala and English music, which incorporates both emotional and rhythmic elements.
- **Problem**: There is a lack of culturally relevant, real-time accessible music systems in Sri Lanka for hearing-impaired students, limiting their emotional and educational engagement with music.

Shortcomings of Existing Systems

- **Existing Systems**: Typically only offer visual feedback or basic haptic responses, without real-time integration of emotional content or cultural relevance.
- **How This Solution Addresses It**: Our system combines AI-driven emotion recognition, SLSL translation, and realistic haptic feedback to ensure hearing-impaired students feel the music, see the lyrics, and understand the emotions behind the song.

APPLICATION OF SPECIALIZATION-BASED KNOWLEDGE

Specialization Areas

- **AI**: Music Emotion Recognition (MER) to classify emotions in music (valence & arousal).
- **NLP**: Real-time translation of lyrics into Sri Lankan Sign Language (SLSL).
- **HCI**: Integration of visual, tactile, and emotional feedback for an immersive experience.

Technologies Used

- **MER Models** (RNN,CNN) for emotion classification
- **Haptic Feedback** from UCI HAR dataset.
- **Web Tech**: React.js (frontend), Flask/FastAPI (backend), MongoDB(storage)

APPLICATION OF SPECIALIZATION-BASED KNOWLEDGE

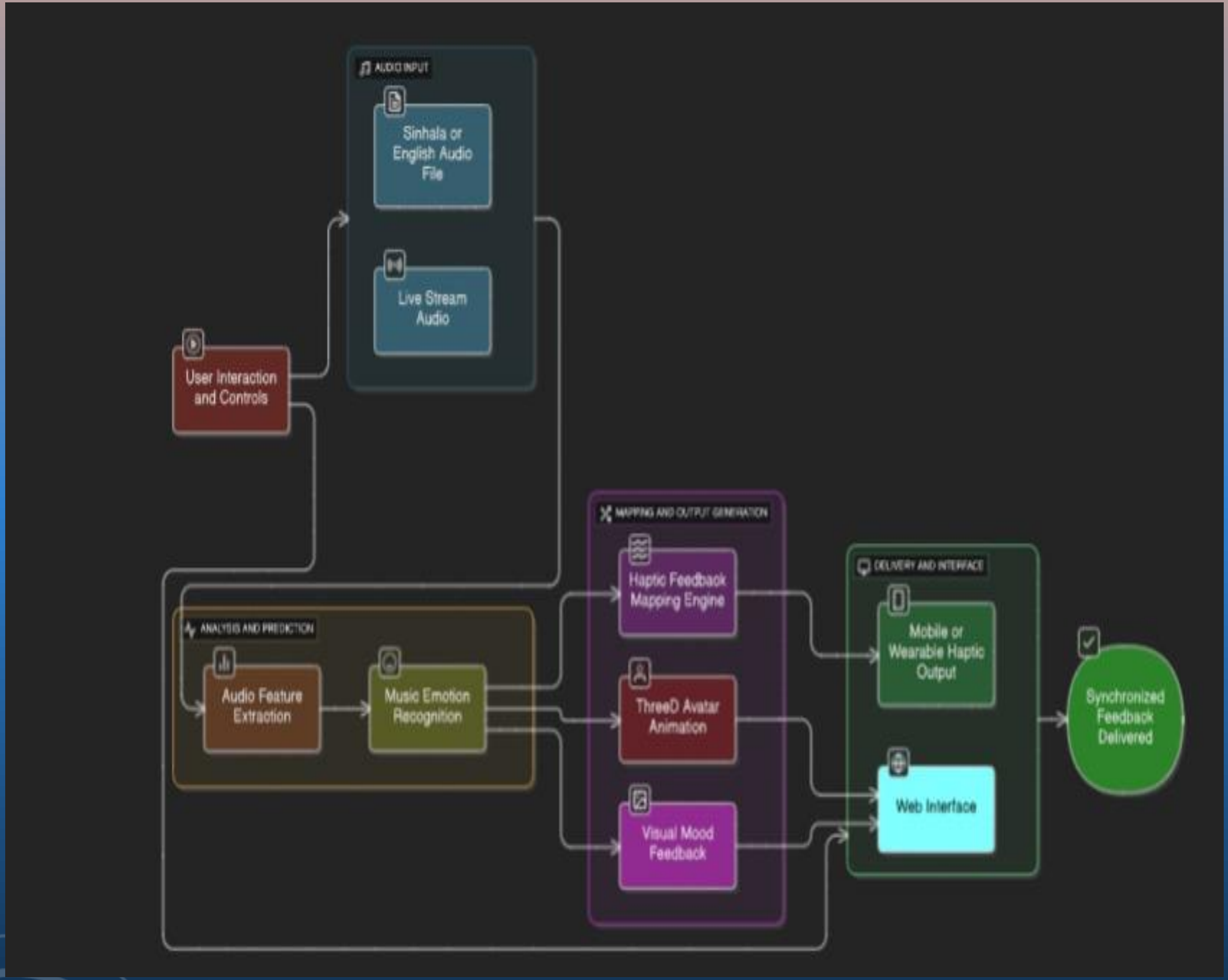
Success Evaluation

- Expected 80% emotion classification accuracy.
- Expected Haptic synchronization within ±50ms.
- User engagement measured through feedback surveys.
- Expected **F1 Score >0.75** for balancing precision and recall.

Data & Ethical Clearance

- Data:
  - **DEAM, EMO-MUSIC**
  - **UCI HAR**
- **Ethical Clearance**: Ensures data privacy, informed user consent, and follows ethical AI guidelines.

Implementation Details



Input:

- **Audio**: Sinhala/English files or live streams.
- **User Preferences**: Haptic intensity, visual feedback, and language choice (SLSL/text).

Process:

- **Feature Extraction**: Tempo, beat, energy, spectral flux.
- **Emotion Recognition**: Predicts emotional tone (valence & arousal).
- **Haptic Feedback**: Converts audio features to vibration patterns.
- **3D Avatar**: Displays sign language and lip movements.
- **Visual Feedback**: Color/animations based on emotion.

Output:

- **Haptic Feedback**: Vibrations on mobile/wearables.
- **Web Interface**: Control settings, avatar, haptic intensity, and visuals.

Real-World Use:

❖ In classrooms, students will use mobile devices or wearables to feel rhythm through haptics, see lyrics in SLSL, and understand emotional content in real-time.

AI-DRIVEN REAL-TIME SLSL E-LEARNING WITH GESTURE CORRECTION & 3D AVATAR STYLES

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INTRODUCTION TO INDIVIDUAL COMPONENT

Aspect	Existing Platforms / Research	Proposed Solution
Proven Gap / Creative Solution	*Hastha*	AI-powered e-learning with real-time SLSL video generation
	EasyTalk	Gesture correction + interactive feedback
Knowledge Gap / Problem Definition	Existing systems focus on American Sign Language (ASL) or static SLSL gestures	AI + NLP + 3D avatars to make SLSL learning interactive and inclusive
Shortcomings of Existing Systems	Pre-recorded or static signs only	avatar styles + feedback for Sri Lankan students

Got the students preference about customized avatar style using google form:

Survey link: https://docs.google.com/forms/d/e/1FAIpQLSeDp6QO9I1_-iU2ch3djpwfQ4FsYn0FGqkUTHm6mB8SQGPcqg/viewform?usp=header

KEY DOMAINS OF KNOWLEDGE

AI & Machine Learning for Gesture recognition, SLSL video generation

Natural Language Processing (NLP) for Sinhala-to-SLSL and SLSL-to-Sinhala translation.

Web Technologies
React (frontend), Flask (backend), MongoDB (data storage)

Computer Vision for Handshape detection, motion tracking, facial expression analysis, gesture error correction

3D Animation & HCI for Avatar design (teacher, realistic, cartoon)

LATEST TECHNOLOGIES USED

Machine Learning & Deep Learning Technologies

- **GANs / Transformers / CNNs / 3D-CNNs** → Sign synthesis & recognition
- **Computer Vision (OpenCV / MediaPipe)** → Handshape, motion & facial expression detection, gesture error correction
- **TensorFlow for** Model training & real-time inference

Supporting Technologies

- **React.js + Flask/FastAPI + MongoDB** → Web platform, APIs & data storage
- **Blender + Three.js** → Customizable 3D avatar rendering (teacher, realistic, cartoon)
- **APIs** → Integration with custom **Sri Lankan Sign Language (SLSL) datasets**

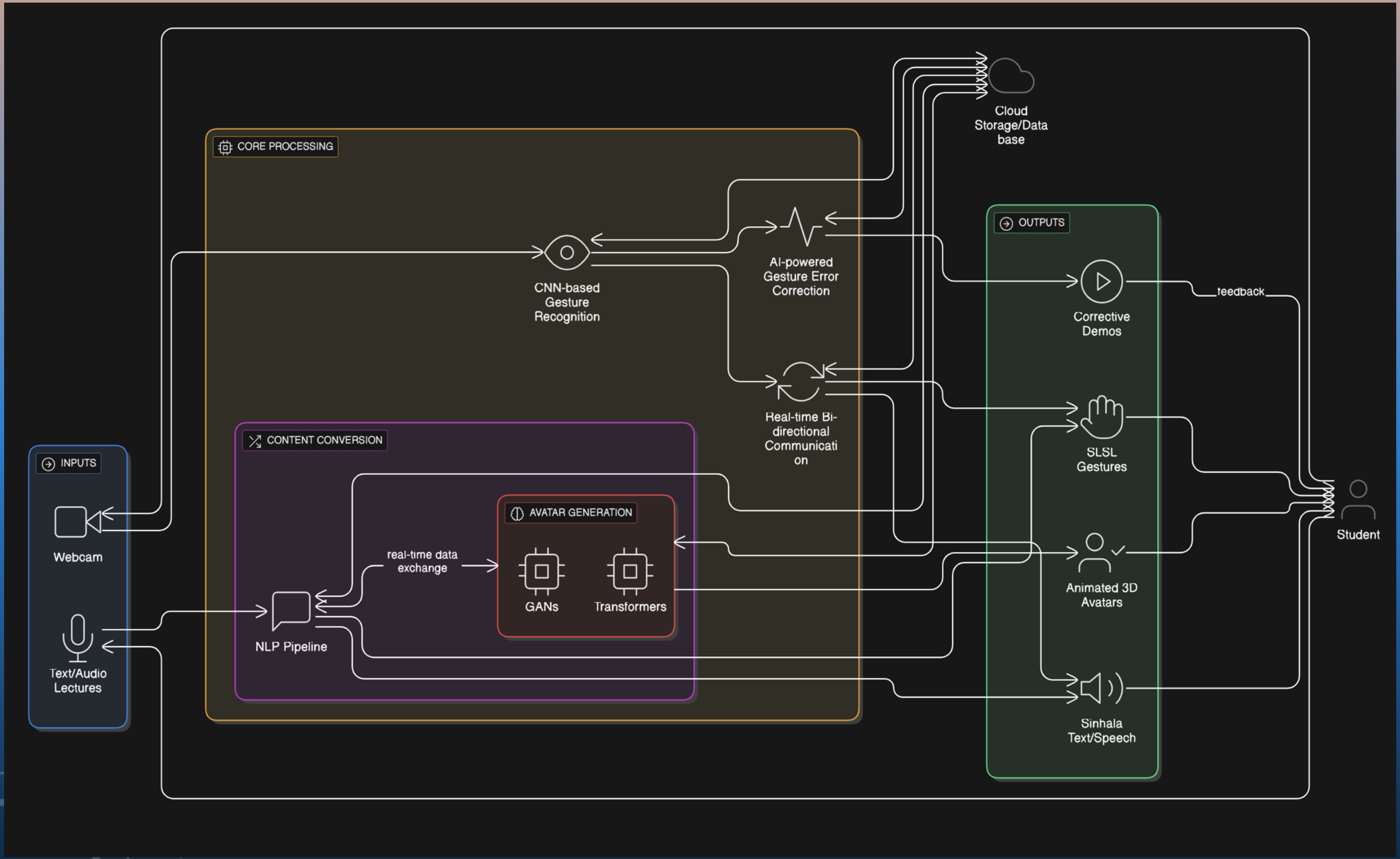
EVALUATION & VALIDATION

- **Metrics**: Precision, Recall, F1-Score (target ≥ 75%).
- **User Testing**: Surveys with educators & hearing-impaired students for usability & satisfaction.
- **Performance**: Real-time response ≤ 500ms latency.

DATA AVAILABILITY & ETHICS

- Public **SLSL datasets** (e.g., SSL400 on Kaggle) .
- No enough data , so currently collecting data.
- Informed consent for data collection from hearing-impaired participants.
- Strict **privacy & anonymization** of video/audio data.
- Compliance with accessibility & inclusivity guidelines (WCAG).

SYSTEM OVERVIEW



REAL - WORLD USAGE

- Assists hearing-impaired individuals in conversations, education, and
- Potential for integration in schools for inclusive learning

AI-POWERED SINHALA LIP-READING ASSISTANT WITH 3D AVATAR FEEDBACK

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KNOWLEDGE GAP

- Lack of Sinhala Visual Speech Recognition (VSR) systems due to absence of large-scale annotated datasets.
- Limited assistive tools for hearing-impaired Sinhala speakers relying on lip-reading in noisy or silent environments.

There are now resources related to the Sinhala language but **Viseme, WAS**, are some of lip-reading systems.

PROBLEM DEFINITION

Existing lip-reading systems focus on English/Chinese; low-resource languages like Sinhala remain unsupported.

SHORTCOMINGS OF EXISTING SYSTEMS

- No localized VSR solutions for the hearing-impaired in Sri Lanka.
- Text-only feedback lacks visual reinforcement.

PROPOSED SOLUTION

Real-time Sinhala lip-reading web app with 3D avatar feedback to enhance understanding.

KEY DOMAINS USED

- Artificial Intelligence (Deep Learning, Computer Vision)
- Natural Language Processing (Phoneme-to-Viseme Mapping)
- 3D Graphics & Animation (Avatar lip-sync)
- Web Technologies (React, Flask, MongoDB)

LATEST TECHNOLOGIES APPLIED

- Transformer-based VSR models (AV-HuBERT, LipNet adaptation)
- Blender/Three.js for real-time 3D lip animation
- TensorFlow for model training

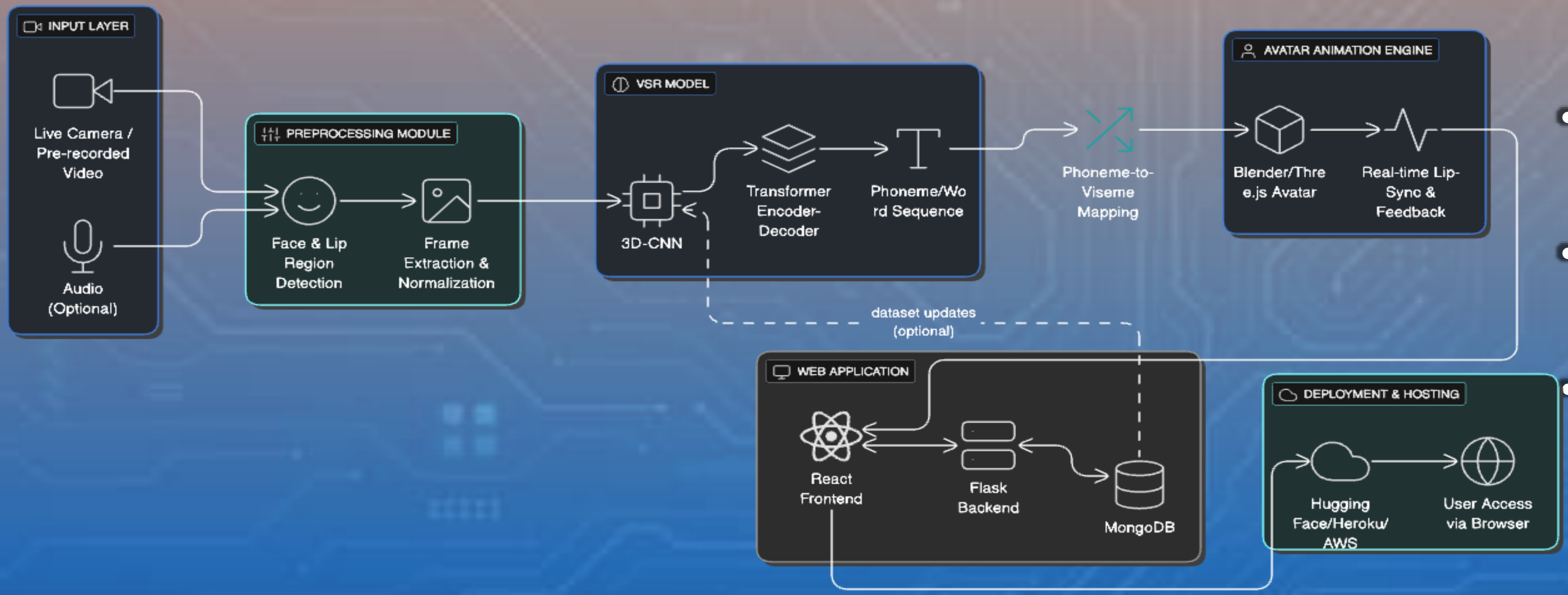
EVALUATION & VALIDATION

- Target Accuracy: ≥75% F1 score for lip-reading recognition
- Real-time performance: <1-second response delay
- User testing with hearing-impaired individuals

DATA AVAILABILITY & ETHICS

- No dataset available to Sinhala language and need to implement a datasets from public video sources (news, education)
- Data anonymization and consent for recorded datasets

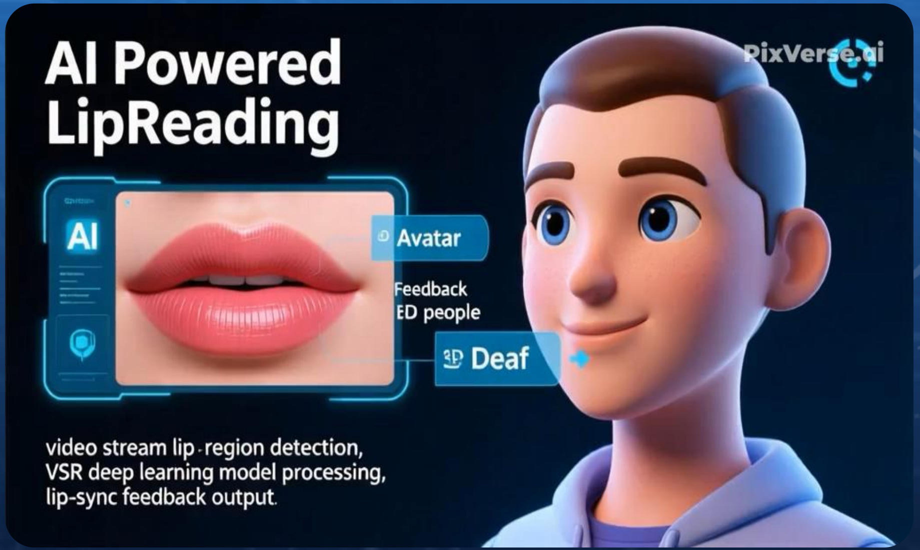
SYSTEM OVERVIEW



- Input: Video stream → Lip region detection
- Processing: VSR deep learning model → Phoneme-to-Viseme mapping
- Output: Text + 3D avatar lip-sync feedback

REAL - WORLD USAGE

- Assists hearing-impaired individuals in conversations, education, and public services
- Used in noisy environments for clearer communication
- Potential for integration in schools, hospitals, and customer service kiosks



EMOTION MONITORING WITH REAL-TIME FEEDBACK AND STAGE-BASED LEARNING OPTIMIZATION

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KNOWLEDGE GAP

- NO EMOTION MONITORING TAILORED FOR SRI LANKAN DEAF LEARNERS.
- SLSL GRAMMATICAL EXPRESSIONS MISREAD AS EMOTIONS.
- LACK OF STAGE-BASED LEARNING OPTIMIZATION.
- NO ANSWERING BEHAVIOR TRACKING (REACTION TIME, CONFIDENCE, ACCURACY).

SHORTCOMINGS OF EXISTING SYSTEMS

- Generic emotion detection → not suitable for SLSL.
- Focus only on emotions, not linked to learning outcomes.
- No adaptive interventions triggered.
- Feedback is generic, not personalized.

PROBLEM DEFINITION

- **Cannot separate emotions from SLSL grammer.**
- **Students disengage in long sessions**
- **Teachers lack insight into confidence & answering behavior.**

PROPOSED SOLUTION

Real-time SLSL e-learning system with emotion monitoring, stage-based optimization, and answering behaviour analysis for adaptive feedback.

KEY DOMAINS OF KNOWLEDGE

Affective Computing (Emotion Monitoring & Behavior Analysis)

Sign Language Processing (SLSL Grammar vs Emotion)

Web Technologies (React, Flask, MongoDB)

Computer Vision (Facial Expression & Sign Input Analysis)

Data Analytics & Visualization (Dashboards & Scores)

LATEST TECHNOLOGIES USED

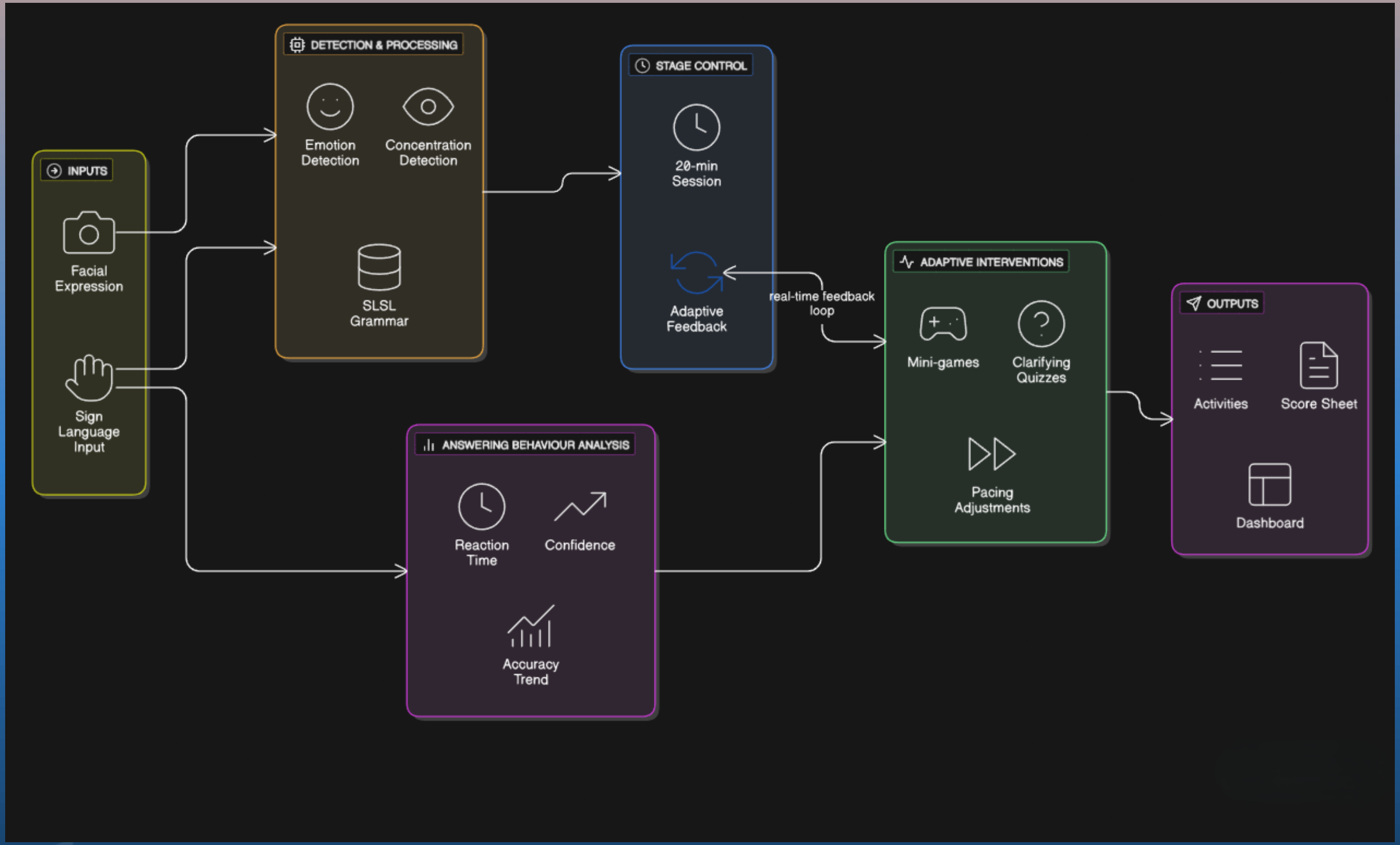
Machine Learning & Deep Learning Technologies

- **Transformer-based Emotion Recognition Models** (FER+, AffectNet).
- **CNN / LSTM Hybrids** for sign-language behaviour tracking.
- **Multimodal Fusion Models** (combining facial and sign inputs).
- **TensorFlow / PyTorch** for training and real-time inference.

Supporting Technologies

- **React.js + Node.js + MongoDB** for web platform & dashboards.
- **OpenCV / MediaPipe** for facial & gesture recognition.
- **Chart.js / D3.js** for performance dashboards.
- **APIs for dataset collection & integration** (custom SLSL datasets).

SYSTEM OVERVIEW



Input:

- Facial expression data + sign language input (student answering behavior).

Processing:

- Emotion & concentration detection.
- Answering behavior analysis (reaction time, confidence, accuracy).
- Stage-based control (20-min sessions with adaptive feedback).

Output:

- Adaptive interventions (mini-games, quizzes, pacing adjustments).
- Score sheet and dashboard for longitudinal performance tracking.

REAL - WORLD USAGE

- Helps teachers monitor **student engagement and confidence** in real-time.
- Provides valuable insights for **schools, e-learning platforms, and special education institutes**.

COMMERCIALIZATION AND OVERALL PROJECT

Potential Market:

- Educational institutions (special schools, universities).
- Healthcare & rehabilitation centers for hearing-impaired users.
- Public service kiosks & customer service platforms.

Revenue Model:

- Freemium web platform with premium enterprise features.
- Licensing to schools, hospitals, and NGOs.
- Paid API access for third-party integrations.

Scalability:

- Can expand to other languages & regions.
- Integration with AR/VR devices for enhanced interaction.

Long-term Vision:

- Commercial product with subscription-based model.
- Continuous AI model improvements with user feedback.